Email Classification Problem and Soln.

The XYZ is an E-commerce company. Daily they receive more than 1000 of mails regarding shipment, product delivery, Quality of product and etc. Issue is every customer mail to one mail id but company have different domains to solve these issues for the customer. As an AI Engineer, you are asked to do redirect the mail from customers complaints mail id to respective department mail id, so that respective department will resolve the issues. Identify the 3 Stages of Problem Domain.

A) Stage1-Machine Learning Stage2-Supervised Stage 3-Regression

B) Stage1-Machine Learning Stage2-Supervised Stage3-Classification

C) Stage1-NLP Stage2-Supervised Stage3-Classification

D) Stage1-NLP Stage2-Unsupervised

Answer is C NLP->Supervised Learning -> Classfication

Stage 1: NLP (Natural Language Processing)

Since we are dealing with emails, they are **unstructured text**. We cannot use standard mathematical algorithms; we have to use NLP to process the language, understand the intent, and convert into machine readable format

Stage 2: Supervised Learning since we know the input(emails) and targeted output (various department to be shipped) this is a good candidate for Supervised learning is the standard approach

Stage 3: Classification since we are going to classify the emails into diff categories like (shipment, QA, ...)